

IT Infrastructure: Are We Building Chaos or Cosmos?



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Agenda



- Change versus continuity: who cares?
- Fragmented IT infrastructure Issues
- Choosing an IT Infrastructure
- Options, Implications, and Conclusions

IT Spending: New Acquisition versus Ongoing Operations



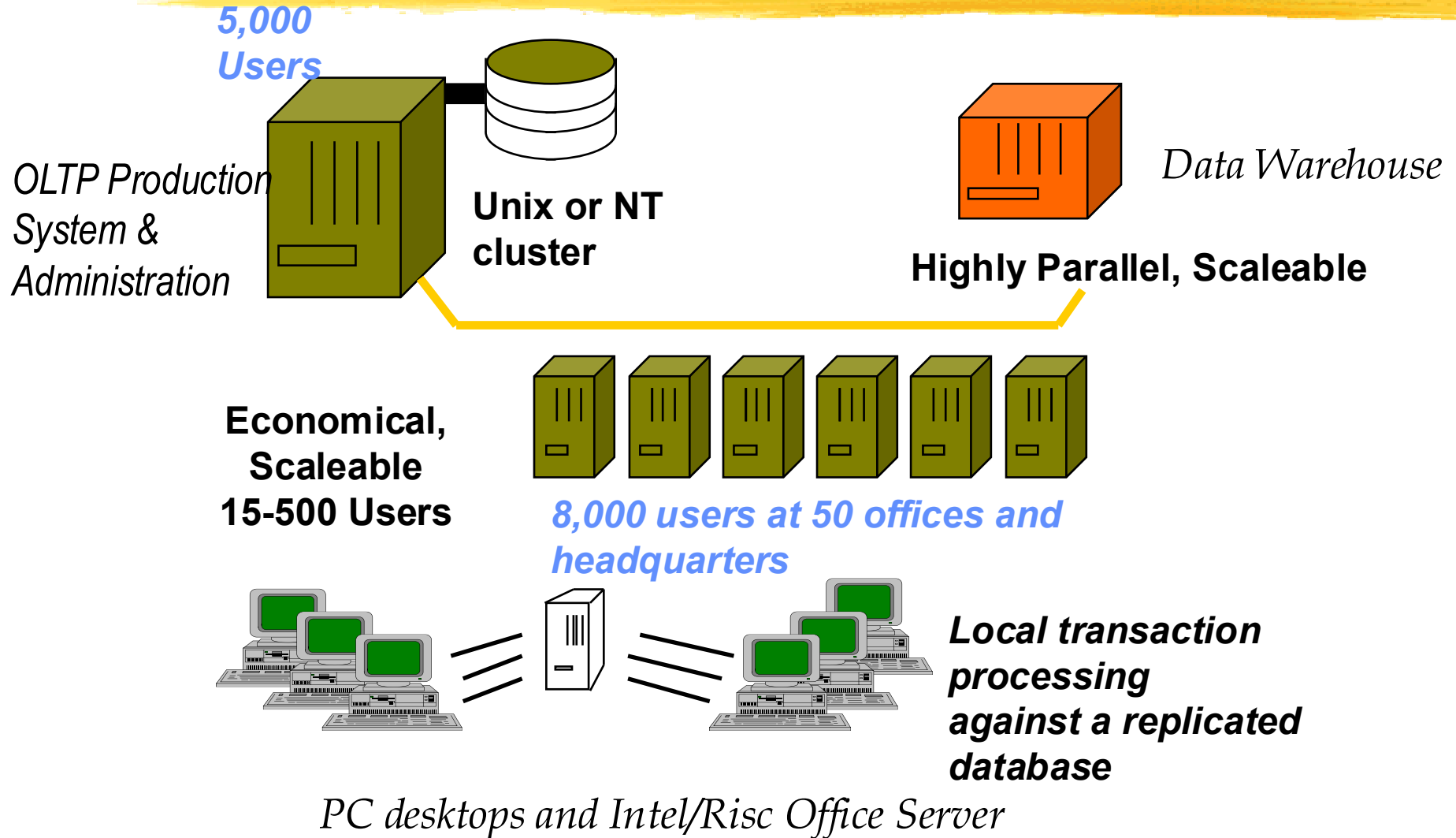
- U.S. enterprises spend $\sim 3\%$ of revenue or value-added on IS
- $\sim 85\%$ of IS spending is for on-going operations — $\sim 15\%$ for new application acquisitions
- CEO looks at 15% for new applications as incremental change & business driver
- CFO and VP IS/CIO look at 85% for on-going operations as area to be closely managed and controlled

IT-Driven Business Imperatives

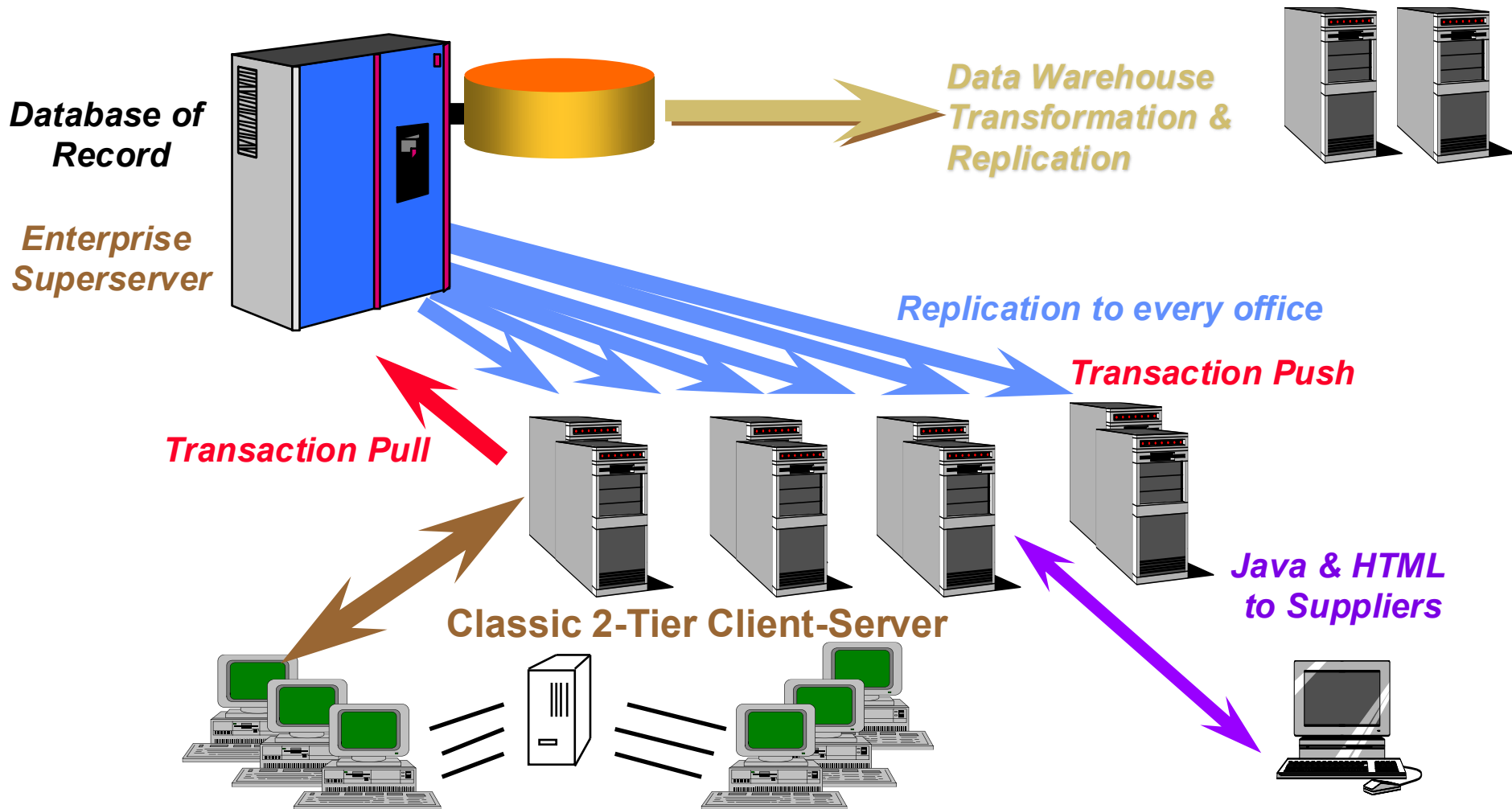


- Increase customer satisfaction
- Empower fewer users
- Increase line access to data
- Increase operational flexibility
- Shorten application development cycle time

Business Necessity Meets IS Reality: Insurance Industry



IS Reality: *Enormous Complexity*



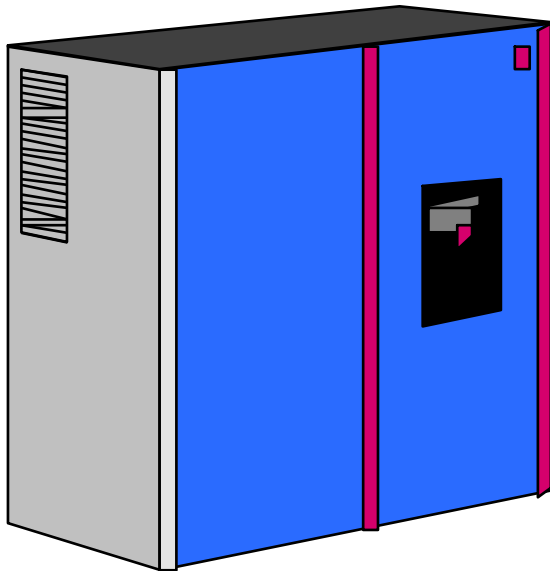
IT Infrastructure Fragments



- Enterprise mainframe
- Geographical islands of automation
- LANs become Intranets
- Data warehousing
- Client-server meets the Web
- E-business is the tip of an iceberg

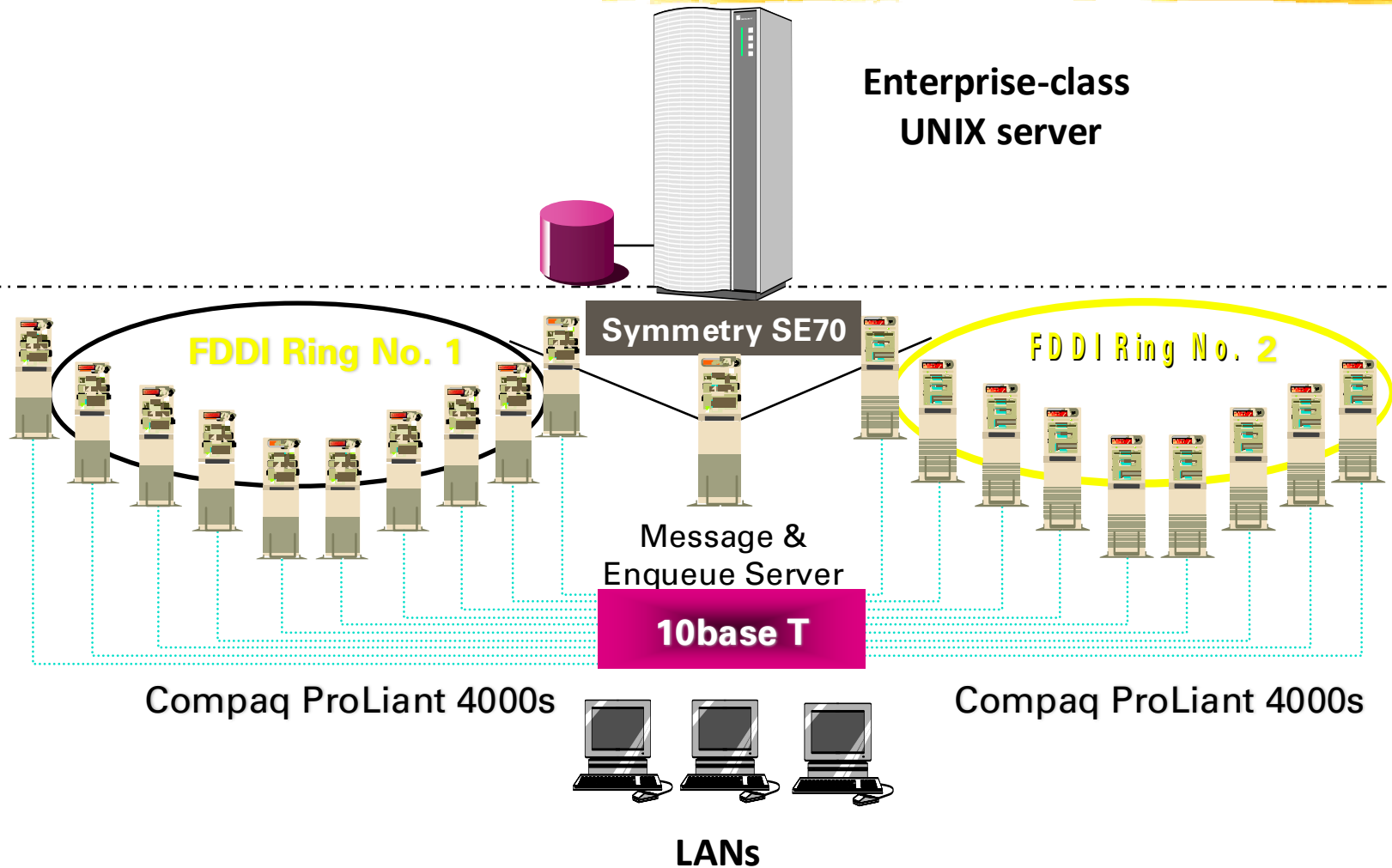
The Devil is in the Technology Details

What Happened to the Mainframe Data Center?

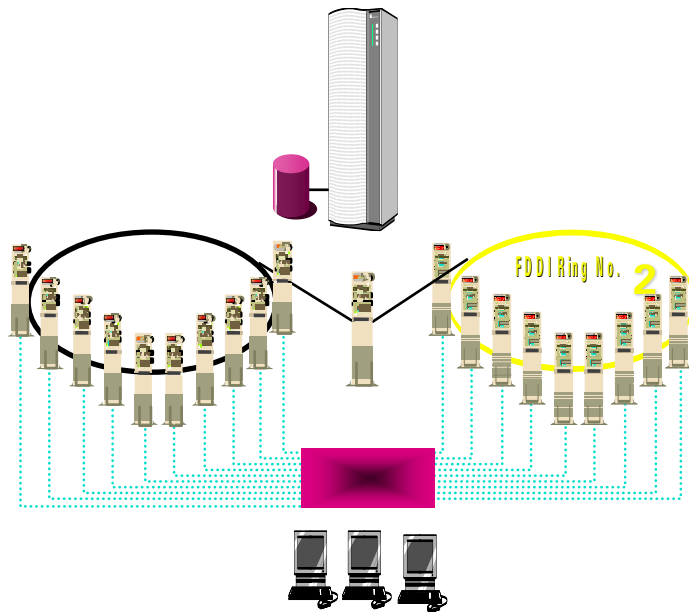


- Thoroughly understood
- Predictable reliability
- Outstanding operational efficiency, productivity, and manageability
- But ...
- Hasn't seen a new ISV application in 7 years
- Is being surrounded or supplanted

New-Breed Enterprise Mainframe

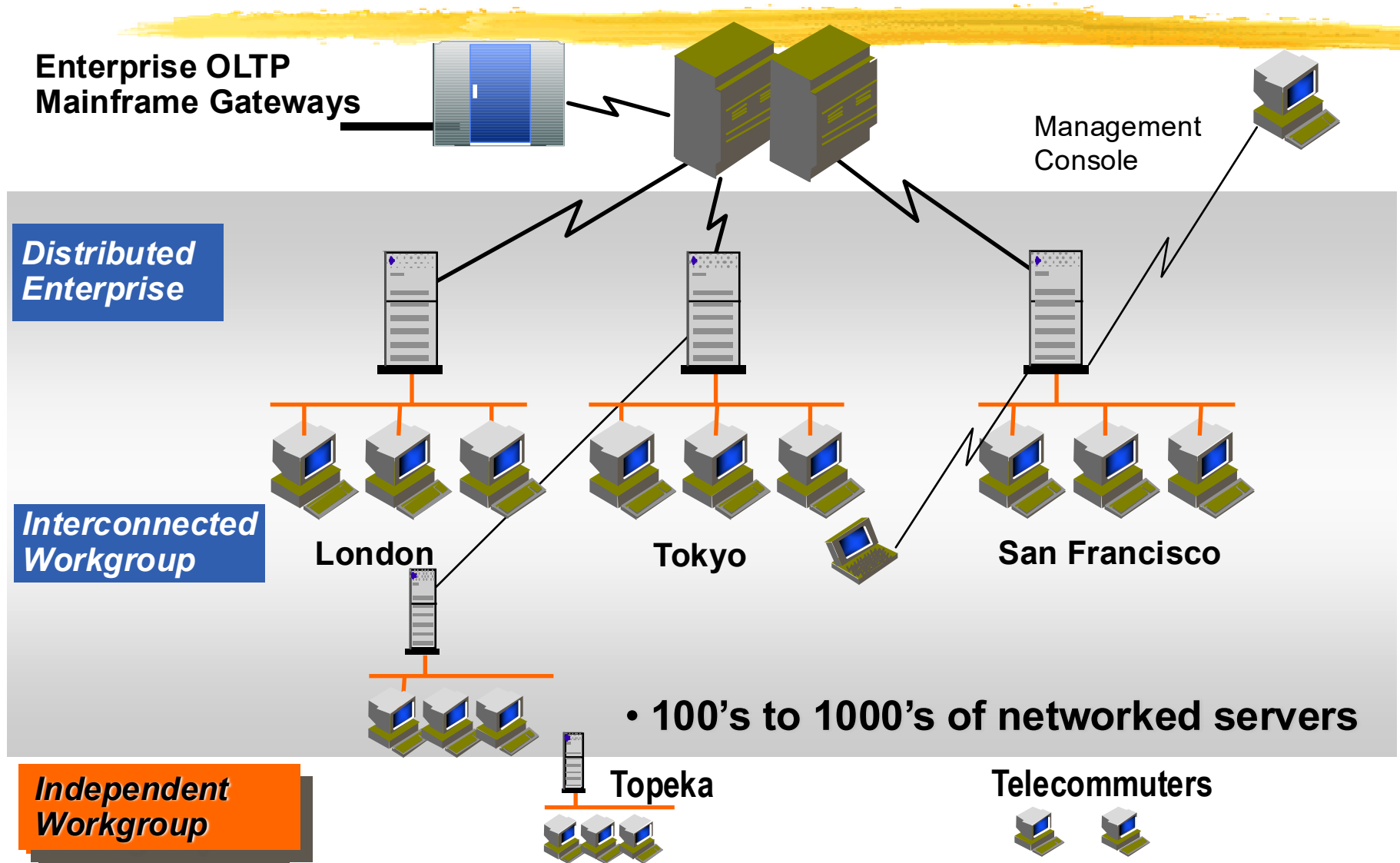


New-Breed Enterprise Mainframe

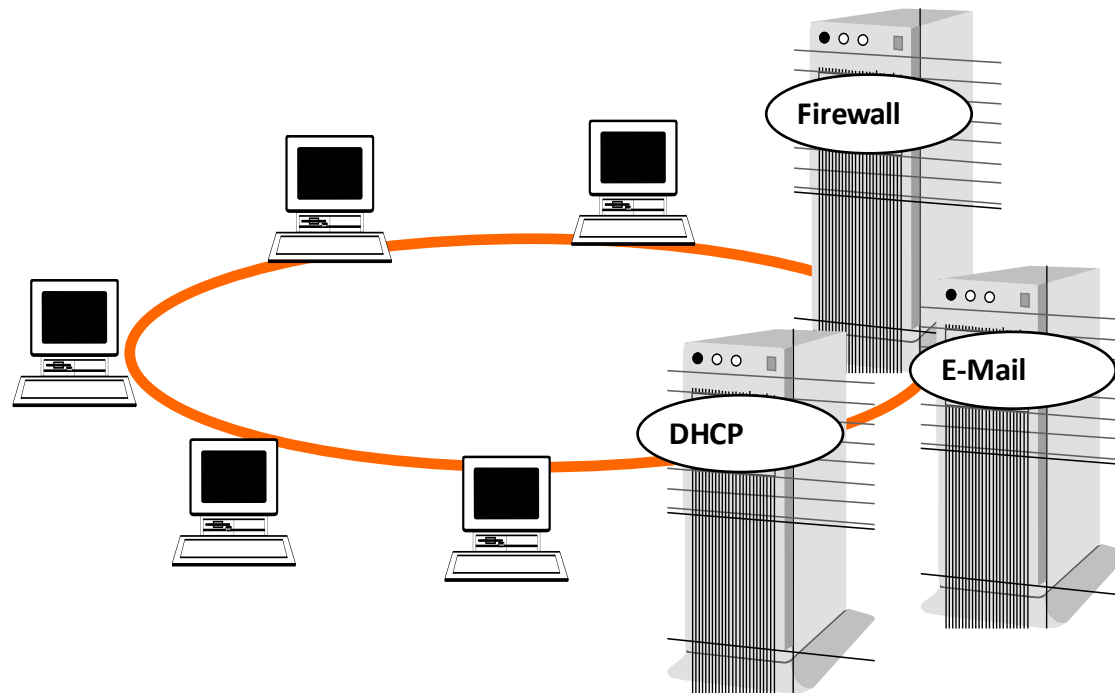


- N-Tier architecture
- Reliability, operability, manageability constraints
- Multi-vendor control issues
- But ...
- Running today's mission-critical enterprise applications: a paradox

Geographically Distributed Enterprise



Shift from NOS to Intranet



New LAN Functions:

- Applications
- Decision support
- Mainframe connectivity
- Exchange server
- Internet proxy
- firewall
- Systems management
- Development
- WINS & DHCP

*“95% of Fortune 1000 Will Have
Intranets By 2000” -- Forrester*

Data Warehousing: “Great Idea, Hard to Implement”



OLTP Transactions

Transformation

Enterprise DW Server

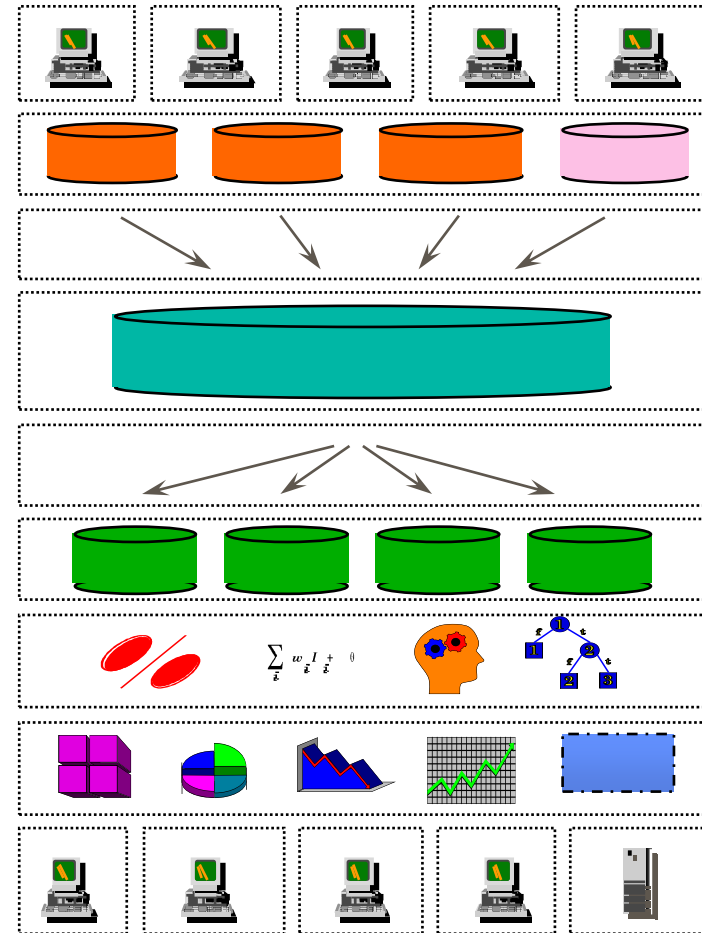
Transformation

Data Mart Servers

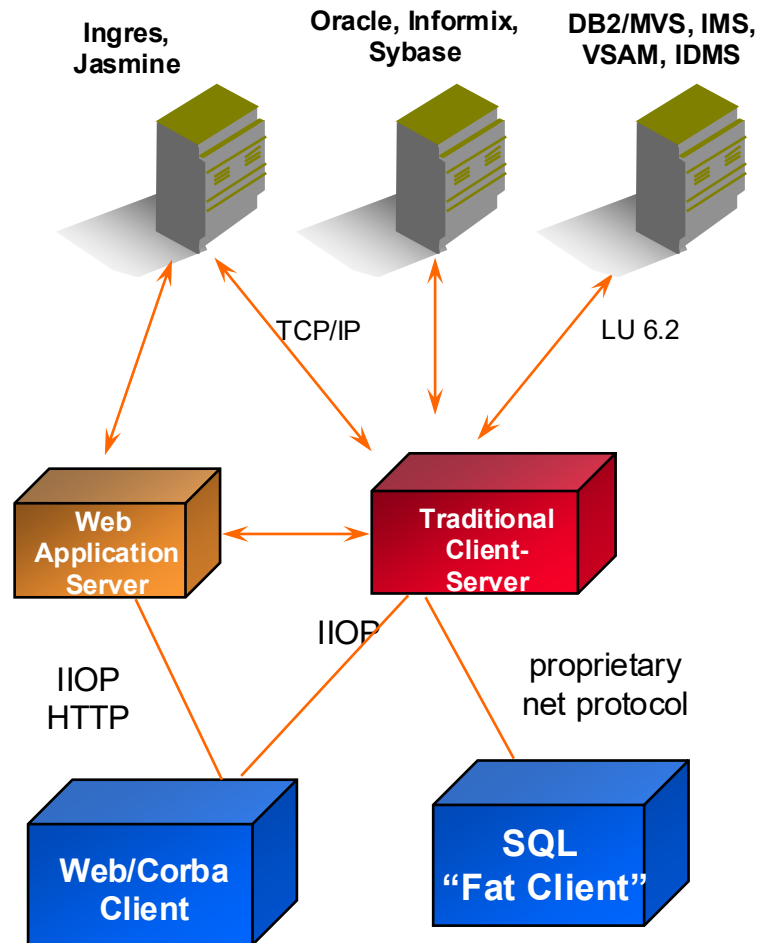
Data Mining Tools

Data Visualization Tools

Decision Support Clients



Client/Server Integration with Web Applications



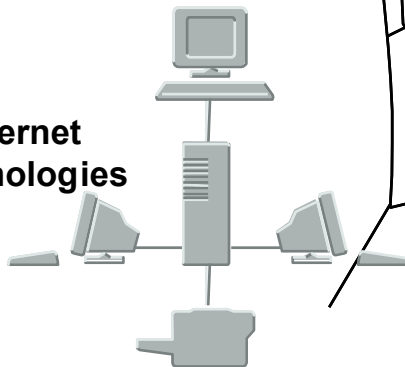
- Just as IS masters client-server technology ...
- Along comes the Web!
- Issues:
 - Legacy C-S integration
 - Evolving but "Not There" standards
 - N-tier complexities
 - Programmer unfamiliarity

CEO's Infinite Resource Planning Vision

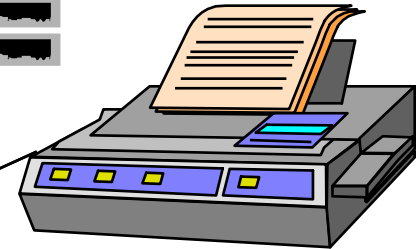


Enterprise Systems

Internet Technologies

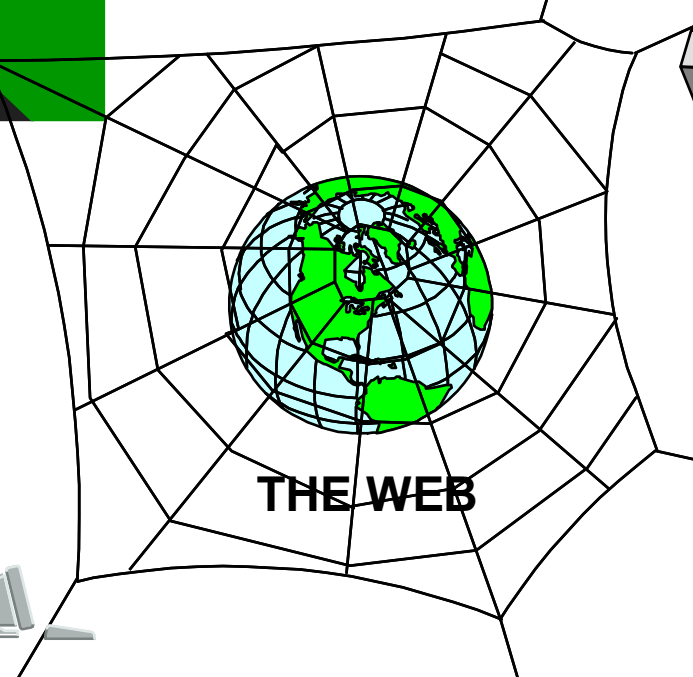
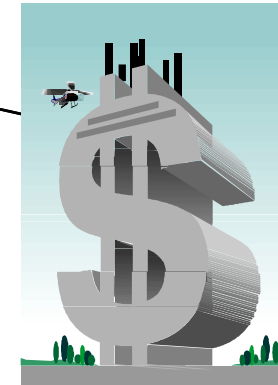


Planning & Forecasting

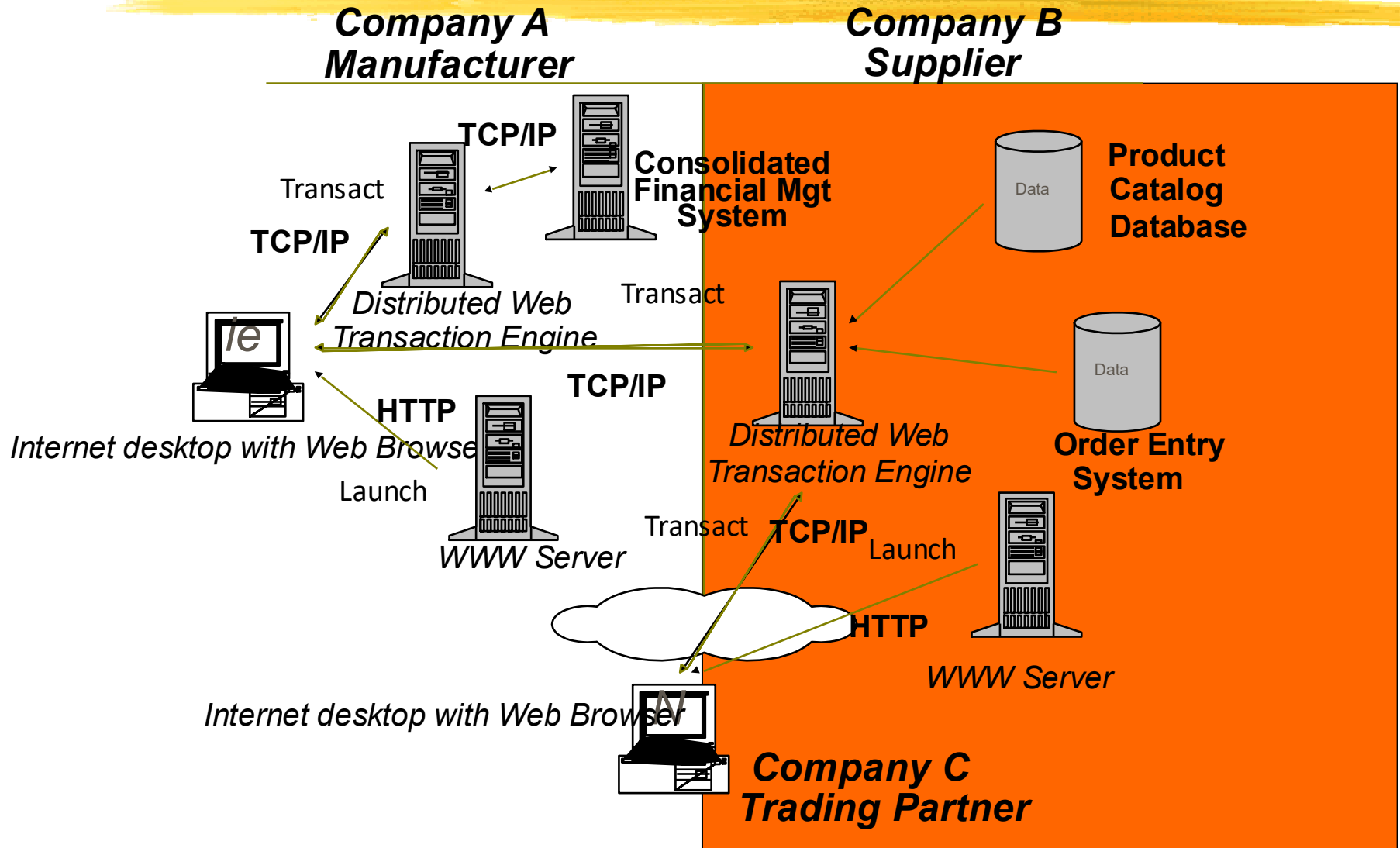


EDI

Commodity Brokerage



Virtual Internet Computing: Reality is Complex, Multi-Company



Building a Common Infrastructure: Management Options

A thick, horizontal yellow brushstroke with a textured, painterly appearance, extending across the width of the slide below the title.

- Do nothing
- Outsource
- Standards-based
- Architecture
- Vendor-based

Do Nothing



- *Take no active steps to build a manageable common IM and EM infrastructure*
- Benefits
 - Just keep on doing what you're doing
- Implications
 - Fragmentation gets worse
 - Ability to disentangle and react diminishes
 - If you buy this, today's speakers have failed abjectly!

Outsource



- *Outside service provider handles IT services*

- Benefits

- Provider assumes future-shock risks
 - Establishes cost-for-service basis
 - One direction for finger pointing

- Implications

- Will provider upgrade your technology base fast enough?
 - What key IS skills will atrophy?
 - Best suited for targeted cost-benefits (e.g., telecomm, “out-tasking”)

Standards-based Infrastructure



- *Using industry standards, create an approved set of hardware, software, and networking technologies*
- Benefits
 - Interoperability yields easier systems integration
 - Less vendor technology lock-in
- Implications
 - Standards come slower; often are incomplete; always behind leading edge
 - Do not always provide the optimal solution
 - IS still has to build/integrate on standard platforms

Architecture



- *Choose the key technologies for an architecture, then the best-of-breed products to implement it*
- Benefits
 - Architecture fits your enterprise's requirements. No "one size fits all" solution
 - Best-of-breed products can change over time while preserving the investment in the architecture
- Implications
 - Requires intense discipline to avoid "exceptions"
 - What is best for the enterprise may not be best for the line-of-business system buyer

Vendor-based



- *Choose a vendor's solutions and build on those*
- Benefits
 - Vendor endures the technology shocks
 - Focused on rolling out leading-edge solutions
 - Some supply value-added technology integration
- Implications
 - No *single* vendor can supply all the infrastructure
 - If the vendor fails, so do you
 - At the mercy of the vendor's technology curve
 - Nevertheless, an attractive choice

Aberdeen's Rating of Management Options



- Do nothing
- Outsource
- Standards-based
- Architecture
- Vendor-based
- Bad idea
- Parts, yes. The whole, never. Staff, never.
- Remember DCE, DME & OSI?
- Good idea. Works best with strong central IS.
- Good idea. Requires excellent relationship and basis for trust.

Aberdeen Conclusions



- IT-driven business is the wave of the present
- Information technology is exploding -- threatening to fragment IS development, operations, and networks into unmanageable pieces
- An architected, vendor-backed *common* infrastructure yields the greatest benefits
 - Must support both information and enterprise management needs